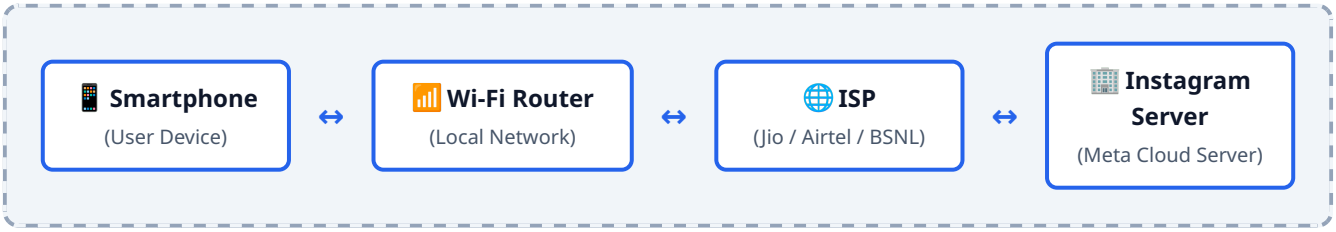
Computer Networking Assignment Answers

Solutions to Questions 1–4

Q1. Connection Diagram: Smartphone to Instagram Server

Below is a diagram showing how a smartphone connects to Instagram servers across various networking components:



Detailed Explanation:

1. **Smartphone (Device):** Sends your request (like opening Instagram feed) over wireless Wi-Fi radio waves.
2. **Wi-Fi Router:** Receives the wireless signals and transmits data packets through optical fiber/Ethernet cables to the internet gateway.
3. **ISP (Internet Service Provider):** Companies like Jio, Airtel, or Fibernet transfer your data across high-speed internet networks.
4. **Instagram Server:** Meta's server processes your request and returns photos/videos back along the same path to your screen.

Q2. 5 Daily Devices & Their Network Types

List of 5 daily devices and the network types they connect to:

Device	Network Type	Connection Details
1. Smartwatch	**PAN** (Personal Area Network)	Connects to a smartphone using Bluetooth over short distances.
2. Laptop	**WLAN / LAN** (Local Area Network)	Connects wirelessly via Wi-Fi or wired Ethernet to a router.
3. Smart TV	**LAN** (Local Area Network)	Connects to home Wi-Fi to stream video content.
4. Wi-Fi Router	**WAN** (Wide Area Network)	Connects local devices to the global Internet provided by the ISP.
5. Smartphone	**WAN / Cellular Network**	Connects to cellular towers (4G/5G) or Wi-Fi networks.

Q3. Difference Between Internet and Intranet

Feature	The Internet	An Intranet
Definition	A global public network connecting computers worldwide.	A private, restricted network for authorized users only.
Access	Public (Accessible by anyone).	Private (Requires authorization/login).
Security	Public traffic with general security.	High security protected behind internal firewalls.

> **College Wi-Fi Example:**
> When connected to College Wi-Fi, opening internal college portals (like attendance or internal exam scores) uses the **Intranet** (restricted to campus). Opening public websites like `google.com` or `youtube.com` uses the **Internet**.

Q4. Data Journey During In-App Search (App: Zomato)

Here is how data travels when searching for food on Zomato:

1. **Request:** When you type *"Pizza"* and press 'Search', your smartphone packages this query into a data packet.
2. **Local Transmission:** The smartphone sends this packet via Wi-Fi or mobile data (4G/5G) to your Wi-Fi router or cellular tower.
3. **ISP Routing:** The ISP routes the packet across global fiber-optic networks directly to Zomato's cloud server.
4. **Server Processing:** Zomato's server queries its database for nearby pizza outlets and sends back a response packet.
5. **Data Arrival:** The response travels back through the ISP to your phone, instantly displaying the restaurant list on your screen.